Team: Markr **Team lead/member:** Guerras Samy Cherif **Repository:** https://github.com/Guerrassc/Markr (public)

1. Approaches compared

Task: turn an assignment brief into a checklist of requirements. All approaches use the synthetic cases in `experiments/m2_extraction`.

	A: regex baseline	B: small LLM	C: larger LLM
Model	No model	`gemini-3.5-flash-lite`	`gemini-3.8-flash` (configured candidate; not run)
Recall	50.3% mean, 10/10 cases	100% mean, 3/10 completed cases	Not measured
Items per brief	3.7	10.7	Not measured
Latency (median / worst)	0 s (local)	31.49 / 186.34 s	Not measured
Estimated cost per action	$0	about $0.00327	about $0.00604
Data needed	None	Brief text + prompt	Brief text + prompt
Failure modes	Misses rules expressed without dates/numbers and some plain-language constraints	7 requests did not complete; latency is high; extraction can still hallucinate	Not tested
Switching effort	None	Low: prompt and provider client are replaceable	Low: same client interface

2. Experiment and results

The script compares regex, small-LLM and large-LLM extraction on 10 synthetic briefs (essay, lab, Haskell, presentation, reflection, French, rubric, email, problem set, thesis). Metric: requirement recall; a labelled requirement counts when one extracted line contains all annotated key strings. Run: `python experiments/m2_extraction/run_experiment.py`; instructions and raw JSON are in the repository.

Results. Regex: 50.3% mean recall, 10/10 cases, 3.7 items/brief. Small LLM: 100% recall on 3/10 completed cases (20 labelled requirements), 10.7 items/brief, 31.49 s median / 186.34 s worst. On those same three cases, regex averaged 45.3%. Seven LLM calls did not complete; large model was not run. Example regex misses (essay): PDF format, Harvard referencing, six-source minimum, title page and late penalty. Evidence is preliminary: synthetic briefs, small sample, imperfect keyword scoring.

3. Chosen approach, limitations and switch trigger

Choice: provisionally use `gemini-3.5-flash-lite`: full recall on the 3 completed cases and lower estimated cost than the larger candidate. Not conclusive: only 3/10 calls completed and worst latency exceeded 3 minutes. **Limitations:** (1) synthetic briefs and imperfect keyword metric; (2) free-tier availability/latency; (3) student brief/draft data leaves Markr, so check provider terms, retention and consent before real use. Scanned-PDF OCR is untested. **Switch if:** recall below 90% on at least 20 varied cases, over 10% failures after retries, unacceptable latency, or unsuitable privacy terms. Evaluate a paid tier, alternative provider or local model; the large model has not been shown to perform better.

4. First architecture sketch

Student + brief/draft → Markr UI/backend (PDF text extraction; prompt in `prompts/`) → **Google Gemini API,** `gemini-3.5-flash-lite` → editable checklist → draft comparison → met/missing/uncertain results. Brief/draft text leaves Markr for Google. API key is stored in local git-ignored `.env`; only placeholders are committed. This is the M2 proposal, not a claim that the complete product exists.

5. First cost estimate

Google Gemini API paid-tier price assumptions recorded 19 Oct 2026: `gemini-3.5-flash-lite` $0.30/1M input, $2.50/1M output; `gemini-3.8-flash` $0.75/1M input, $3.75/1M output. Source: https://ai.google.dev/gemini-api/docs/pricing; verify current price/model availability before deployment. Small-model experiment total: 591 input / 430 output tokens across 3 successful calls = about 197 / 143 per extraction; measured extraction cost is $(197 \times 0.30 + 143 \times 2.50)/10^6 = \$0.00042$ per brief.

End-to-end estimate (not measured): 3,800 input + 850 output tokens per brief-and-draft action. Small: $0.00327/*action*, or $9.80/*month* at 3,000 actions. Large: $0.00604/*action*, or $18.11/*month*. Excludes hosting/storage/tax; development calls used free tier.

6. Plan to MVP (due Monday 16 November 2026)

Date	Task	Owner	Done when
20–22 Oct	Re-run/extend model comparison; document failures and final model decision	Samy	Reproducible results and updated table
23–28 Oct	PDF/text upload and text extraction; connect chosen model end to end	Samy	5 sample briefs processed without mocked responses
29 Oct–5 Nov	Draft-versus-checklist comparison and simple browser UI	Samy	Full upload-to-result flow works
6–9 Nov	Test with real briefs only with informed consent; record bugs and failure cases	Samy	Test notes and at least 3 known issues
10–13 Nov	Update README, architecture, `.env.example`; test on another laptop	Samy	Setup in 5 steps or fewer on a clean machine
14–15 Nov	Record 3-minute demo; prepare one-page M3 summary and submit	Samy	Video link, bug list and tagged submission

M2 status: required artefacts are present; LLM evidence is partial (3/10 calls completed), and the larger model is untested.